

Numerical Methods

Image Retrieval, Image Forensic, Secret Image Sharing

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JPEG Encoder and Decoder

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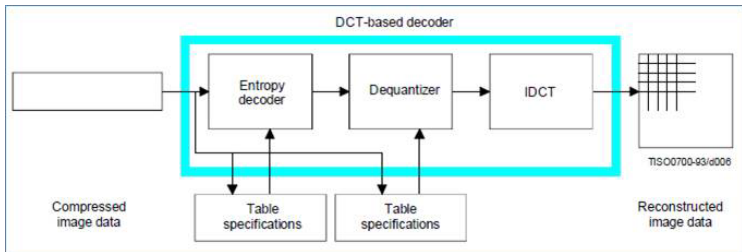
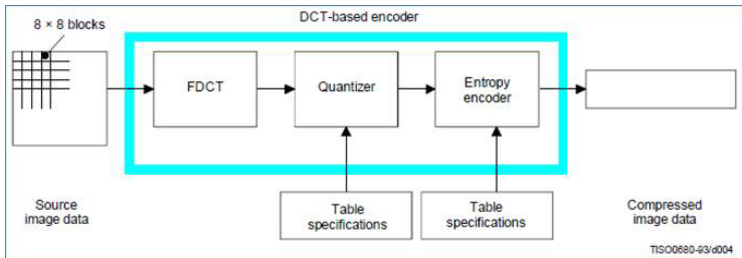
GGD Model

Moment
Estimation

Max-Like
Estimation

Lena Results

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Statistical Models of DCT Coefficients

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- It is generally believed that the luminance DC coefficients 亮度直流系数 may be modeled as Gaussian distribution and the luminance AC coefficients 亮度交流系数 as generalized Gaussian distribution 广义高斯分布.
- It is demonstrated that the chrominance DCT coefficients 色度 DCT 系数 exhibit the same distributions as for the luminance component in **YCbCr** color space 颜色空间.
- The widths of the DCT coefficient distributions shrink with the frequency values increasing. That is, the distributions of higher frequency coefficients have smaller variance, indicating a stronger peak zeros or generally smaller values.



Histogram of DCT coefficients of Bridge

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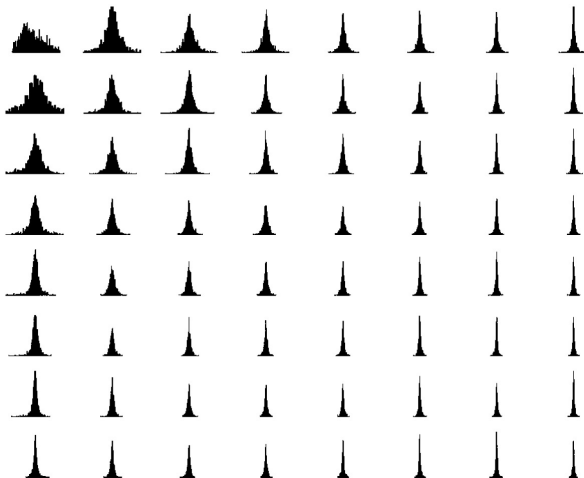
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Generalized Gaussian Distribution

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- The probability density function (PDF) 概率密度函数 of GGD model is defined as

$$f(x; \alpha, \beta) = \frac{\beta}{2\alpha\Gamma(1/\beta)} \exp \left\{ - \left(\frac{|x|}{\alpha} \right)^\beta \right\}$$

where $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$ is the Euler Gamma function.

- This PDF is a two-sided symmetric density with two distributional parameters β 形状因子 and α 方差因子 that control the shape and standard deviation of the density, respectively.
 - With $\beta = 2$, $\alpha = \sqrt{2}$, it becomes a standard normal distribution;
 - As $\beta \rightarrow \infty$, it approximates the uniform distribution;
 - By setting $\beta = 1$, $\alpha = 1/\lambda$, the Laplacian distribution is obtained.



Mallat's Moment Estimation

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- The distributional parameters α and β may be computed by measuring the first and second moment of the GGD distribution.

$$m_1 = \int |x| f(x; \alpha, \beta) dx = \frac{\alpha \Gamma(2/\beta)}{\Gamma(1/\beta)}$$

$$m_2 = \int x^2 f(x; \alpha, \beta) dx = \frac{\alpha^2 \Gamma(3/\beta)}{\Gamma(1/\beta)}$$

- From the above two equations, we can derive that

$$\frac{m_1^2}{m_2} = \frac{\Gamma^2(2/\beta)}{\Gamma(1/\beta) \Gamma(3/\beta)}$$

- Given the samples x_i (N denotes the number of samples), m_1 and m_2 can be estimated by

$$\hat{m}_1 = \frac{1}{N} \sum |x_i|, \quad \hat{m}_2 = \frac{1}{N} \sum x_i^2$$



GGD Maximum-Likelihood Estimator

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- Let us define the likelihood function 似然函数 as

$$L(x; \alpha, \beta) = \log \prod_{i=1}^N p(x_i; \alpha, \beta).$$

- From the theory of parameter estimation, the ML estimates of α and β are defined as

$$(\hat{\alpha}, \hat{\beta}) = \underset{\alpha, \beta}{\operatorname{argmax}} L(x; \alpha, \beta)$$

- Then $\hat{\alpha}$ and $\hat{\beta}$ can be obtained by resolving the following two likelihood equations 似然方程:

$$\frac{\partial L(x; \alpha, \beta)}{\partial \alpha} = -\frac{N}{\alpha} + \sum_{i=1}^N \frac{\beta |x_i|^\beta \alpha^{-\beta}}{\alpha} = 0$$

$$\frac{\partial L(x; \alpha, \beta)}{\partial \beta} = \frac{N}{\beta} + \frac{N\Psi(1/\beta)}{\beta^2} - \sum_{i=1}^N \left(\frac{|x_i|}{\alpha} \right)^\beta \log \left(\frac{|x_i|}{\alpha} \right) = 0$$



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- According to the first likelihood equation, we have

$$\hat{\alpha} = \left(\frac{\beta}{N} \sum_{i=1}^N |x_i|^\beta \right)^{1/\beta}.$$

- By inserting $\hat{\alpha}$ into the second likelihood equation, we get the following equation about the shape parameter β .

$$1 + \frac{\Psi(1/\hat{\beta})}{\hat{\beta}} - \frac{\sum_{i=1}^N |x_i|^{\hat{\beta}} \log |x_i|}{\sum_{i=1}^N |x_i|^{\hat{\beta}}} + \frac{\log \left(\frac{\hat{\beta}}{N} \sum_{i=1}^N |x_i|^{\hat{\beta}} \right)}{\hat{\beta}} = 0$$

where $\Psi(\cdot)$ is Digamma function defined by
 $\Psi(x) = \Gamma'(x)/\Gamma(x)$.



GGD Parameters of Lena 8×8 block-DCT Coefficients

- Mallat's Moments Estimation (UP); Maximum-Likelihood Estimation (DOWN)

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Moment Estimation

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Lena Results

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	0	1	2	3	4	5	6	7
0	-	(11.0,0.54)	(2.29,0.46)	(1.18,0.45)	(0.79,0.45)	(0.80,0.49)	(1.17,0.58)	(1.80,0.73)
1	(3.02,0.45)	(2.05,0.45)	(0.98,0.43)	(0.74,0.43)	(0.93,0.50)	(0.93,0.53)	(1.22,0.61)	(2.10,0.82)
2	(0.92,0.43)	(0.80,0.42)	(0.58,0.41)	(0.72,0.45)	(0.70,0.48)	(0.80,0.53)	(2.03,0.78)	(2.38,0.92)
3	(0.72,0.46)	(0.78,0.47)	(0.79,0.48)	(0.81,0.50)	(0.85,0.53)	(1.24,0.63)	(1.62,0.74)	(2.53,1.03)
4	(1.34,0.60)	(1.43,0.62)	(1.59,0.64)	(1.18,0.59)	(1.23,0.63)	(1.58,0.74)	(2.25,0.96)	(2.20,1.00)
5	(2.29,0.83)	(1.93,0.78)	(1.85,0.76)	(1.96,0.79)	(1.97,0.84)	(2.14,0.94)	(2.31,1.04)	(2.25,1.07)
6	(2.40,0.99)	(2.50,1.02)	(2.39,1.03)	(2.14,0.95)	(2.20,0.98)	(2.59,1.17)	(2.78,1.33)	(2.54,1.27)
7	(2.78,1.20)	(2.73,1.21)	(2.72,1.27)	(2.57,1.22)	(2.44,1.17)	(2.56,1.25)	(2.56,1.32)	(2.62,1.41)

	0	1	2	3	4	5	6	7
0	-	(6.49,0.47)	(1.82,0.44)	(2.18,0.52)	(1.77,0.55)	(2.06,0.64)	(2.28,0.75)	(2.71,0.89)
1	(3.41,0.47)	(1.41,0.42)	(1.18,0.44)	(1.34,0.49)	(1.77,0.59)	(1.93,0.67)	(2.26,0.78)	(2.69,0.94)
2	(1.61,0.49)	(1.04,0.45)	(1.02,0.46)	(1.44,0.53)	(1.66,0.61)	(1.74,0.67)	(2.66,0.90)	(2.82,1.03)
3	(1.85,0.59)	(1.64,0.57)	(1.50,0.56)	(1.49,0.59)	(1.85,0.67)	(2.08,0.77)	(2.32,0.89)	(2.73,1.10)
4	(2.47,0.76)	(2.15,0.71)	(2.22,0.72)	(1.81,0.68)	(1.90,0.75)	(2.33,0.90)	(2.58,1.05)	(2.49,1.10)
5	(2.90,0.96)	(2.49,0.89)	(2.47,0.88)	(2.46,0.90)	(2.38,0.93)	(2.50,1.05)	(2.56,1.12)	(2.50,1.17)
6	(2.67,1.07)	(2.78,1.11)	(2.63,1.10)	(2.48,1.05)	(2.45,1.06)	(2.70,1.22)	(2.81,1.35)	(2.59,1.30)
7	(2.86,1.23)	(2.79,1.24)	(2.78,1.30)	(2.66,1.27)	(2.55,1.22)	(2.62,1.28)	(2.59,1.34)	(2.62,1.41)



A Simple (k, n) Threshold Scheme

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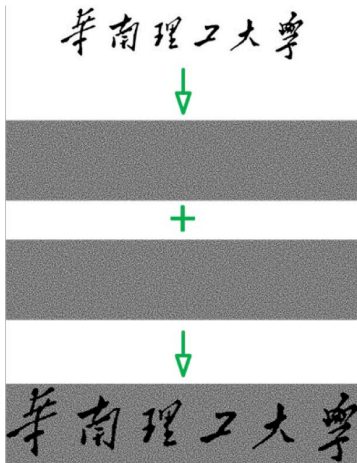
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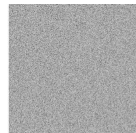
Max-Like
Estimation

Lena Results

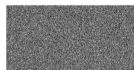
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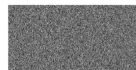
(a)



(b)



(c)



(d)



(e)



(f)



(g)



A Simple (k, n) Threshold Scheme

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- Given k points in the 2-dimensional plane $(x_1, y_1), (x_2, y_2), \dots, (x_k, y_k)$ with distinct x_i 's, there is one and only one polynomial $q(x)$ of degree $k - 1$ such that $q(x_i) = y_i$ for all i .
- Without loss of generality, we can assume that the data D is (or can be made) a number. To divide it into pieces D_i , we pick a random $k - 1$ degree polynomial $q(x) = a_0 + a_1x + \dots + a_{k-1}x^{k-1}$ in which $a_0 = D$.
- Evaluate $D_1 = q(1), D_2 = q(2), \dots, D_n = q(n)$.
- Given any subset of k of these D_i values (together with their identifying indices), we can find the coefficients of $q(x)$ by interpolation, and then evaluate $D = q(0)$.
- Knowledge of just $k - 1$ of these values, on the other hand, does not suffice in order to calculate D .